

Valeria Ramirez Osorio

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HIGHLIGHT OF QUALIFICATIONS

- **Eligible for Masters 4-16 month Co-Op placement starting Sept. 2026.** Flexible for extension
- **Scope & solve complex challenges**, designing and implementing scalable solutions to ambiguous problems using strong problem-solving, algorithms, and debugging skills.
- **Full-stack development lifecycle**, building end-to-end applications from system design to implementation and testing using modern frameworks, RESTful APIs, and automated testing practices.
- **Data management & analysis**, working with structured and unstructured data using SQL and Python to design efficient data pipelines, perform analysis, and support data-driven decisions.
- **User-centered development**, incorporating accessibility (WCAG) and usability principles into applications, informed by user feedback to create intuitive and inclusive experiences.

EDUCATION

McMaster University — Hamilton ON

Computing and Software, Co-Op

Masters of Engineering

2025 - 2026

University of Toronto — Toronto ON

Computer Science Specialist and Mathematics Minor

Honours Bachelor of Science

2020 - 2025

- High Distinction
- Dean's List (2021, 2022, 2023, 2024)

SKILLS HIGHLIGHTS

Software

- Python, Java, C++, CI/CD Tools, Bash, UNIX/Linux, Microsoft Products, Jupyter Notebook, Jira, Slack, Miro, Google Tools, Automation, Figma, Generative AI, Agentic AI, Prompt Engineering, LLMs, Machine Learning
- Full-Stack Development, APIs & System Integration, Software Testing & Debugging, Agile & Scrum

Data Science

- SQL, MySQL, R, Excel, GCP, Docker, Kubernetes, Data Visualization tools
- Data Interpretation and Manipulation, EDA, Data processing and Analytics, Statistics, Probability

Design

- UI & UX, HCI, User-Centered Design, Interaction and Interface Design Thinking, Lean Startup, Accessibility
- Frontend UI (HTML, CSS, JavaScript, React, NodeJS), Prototyping, Visual Design, AutoCAD, Latex, Confluence

Languages

- English (Fluent), Spanish (Fluent), French (Intermediate)

RELEVANT EXPERIENCE

Summer Student Engineering Internship

May 2021 – August 2024

Bell

- Led annual projects aimed at enhancing team efficiency, including application development, inventory database implementation, and software testing/debugging through Python, Java, JavaScript, HTML, SQL, etc.
- Contributed to team success, such as 5G Slicing, RAN, and SRE, through collaboration, effective communication, in Agile and Scrum practices. Used Jira and Confluence to manage the SDLC and ensure timely project delivery.
- Applied analytical and problem-solving skills, adaptability, and initiative to provide innovative solutions, manage priorities, and support a productive team environment.
- Developed an automation application integrating XCAL software with Python and JavaScript, enabling seamless interaction with a Slack channel. Users could send commands and receive real-time test results, allowing thousands of cellular tests to run on physical hardware without requiring continuous monitoring. This significantly reduced manual oversight and increased testing efficiency.

Application and Database Developer

September 2024 – August 2025

University of Toronto

- Independently created and implemented a website's front-end and back-end, leveraging full creative freedom to deliver a unique design and functionality for integration into the Department of Historical Studies.
- Utilized SQL, Oracle, Omeka, React, Node.js, JavaScript, HTML, and additional tools to design, develop, and deploy the platform.
- Coordinated closely with the project supervisor by providing clear progress updates, gathering feedback, and translating requirements into actionable technical tasks to ensure alignment with departmental expectations.
- Led the project from conception to launch with minimal supervision, demonstrating strong critical-thinking skills, effective time management, and the ability to meet all development milestones within the established timeline.

Human-AI Research Internship

May 2026 – July 2026

Graz University of Technology

Graz, Austria

- Conducted research on synergetic human-AI task performance and collaborative decision-making systems.
- Developed and evaluated agentic AI workflows for adaptive, human-centered applications.
- Collaborated with interdisciplinary teams under expert supervision in trustworthy AI research.
- Strengthened skills in communication, teamwork, critical thinking, and research-driven problem solving.

RECENT PROJECTS

Development of an Educational AI-Driven Chatbot | *Master's Research*

September 2025 – Present

- Designing and developing an interactive chatbot application powered by generative AI to simulate professional client interactions and enhance engineering students' requirement elicitation skills.
- Implementing frontend architecture and user experience design to create realistic, classroom-applicable client communication scenarios.
- Collaborating with a research supervisor and graduate peers to iterate towards a functional prototype and prepare for a formal user study evaluating educational impact.

Understanding the Impact of Using Generative AI Tools in a Database Course | May 2024 – August 2024

- Manipulated and organized quantitative and qualitative data using Python and Microsoft Excel to extend trends for computer science professor to understand interaction of students and GenAI in database courses.

Multiple Representations in Introductory Programming: A Pilot Study | September 2024 – January 2025

- Collaborated with a PhD student and three professors on a pilot study examining how multiple visualizations support learning in introductory programming. Contributed to the design of interactive exercises for first-year computer science courses.
- Developed backend functionality for interactive Python-learning exercises using JavaScript, Python, and HTML, featuring synchronized views of code execution, memory state, and real-world analogies.

Understanding LLM Prompting for Responsible Student Use Study | September 2025 – December 2025

- Conducted an independent research study on how prompt design affects the performance, accuracy, and educational usefulness of large language models, including ChatGPT, Gemini, and Claude.
- Designed evaluation frameworks and Python-based analysis pipelines to assess model outputs across varied prompt conditions, presenting findings on readability, correctness, and educational trade-offs.

ADDITIONAL EXPERIENCE

Teaching Assistant | *McMaster University*

September 2025 – Present

- Facilitated engaging and interactive tutorials, providing clear explanations and fostering student participation through real-world examples, collaborative problem-solving, and class discussions.
- Hosted regular office hours, offering personalized academic assistance and troubleshooting difficulties related to course material, helping students bridge gaps in understanding.
- Adapted to a new teaching environment by quickly learning and mastering digital teaching tools and online course platform CalcCheck. Courses Include: *Logical Reasoning for Computer Science*, and *Algorithm Design and Analysis*

Computer Science and Mathematics Tutor | *Self-employed*

September 2024 – Present

- Crafted lessons and practices for high school students to improve their understanding and knowledge.
- Focused on Advanced Functions, Calculus, Computer Science, Languages: Java and Python
- Experienced in supporting students with diverse learning needs, including those with dyslexia, and ADHD.